Infant Statistical Speech Segmentation

Eight-month-olds can use transitional probabilities in a brief artificial-speech stream to distinguish word-like units from part-words; this is laboratory segmentation, not proof that statistics alone explain natural language.

GRADE: HIGH-LAB; LOW-TRANSFER | STATUS: STABLE | DOMAIN: LANGUAGE, LEARNING | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Fluent speech does not arrive pre-segmented. In a classic laboratory demonstration, 8-month-old infants exposed to two minutes of a continuous artificial-speech stream later distinguished syllable sequences that had been “words” in that stream from sequences that straddled word boundaries. Transitional probability was the available cue.

EVIDENCE AND CAUSAL ASSESSMENT

Saffran, Aslin & Newport (1996), Science: 8-month-olds; familiarization with a continuous stream of three-syllable nonsense words; test preference for words vs part-words. Aslin, Saffran & Newport (1998), Psychological Science, held syllable frequency constant and still found discrimination based on conditional probability — a conceptual replication of the computational claim, not a household-vocabulary study. The design manipulates the statistics of the input and measures immediate segmentation. It does not assign meaning, social reference, or later vocabulary.

EFFECT SIZE AND PERSISTENCE

The result is immediate looking/listening preference after a short exposure. Transfer to the child’s language, durability without further exposure, and contribution to real vocabulary are not established by these experiments.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a. Artificial language; small classic samples; laboratory attention. Adjacent Chapter 6 contingency learning is a different action–outcome claim.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean the infant already knows words in the adult sense, or that a parent must provide a statistically engineered stream. Detecting a regularity is not knowing a meaning.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary talk in a shared world is consistent with the finding. Segmentation software for infants is not.

VERIFIED SOURCES & DOI IDENTIFIERS

- Saffran, Aslin & Newport (1996), Science 274:1926–1928
- Aslin, Saffran & Newport (1998), Psychological Science 9:321–324